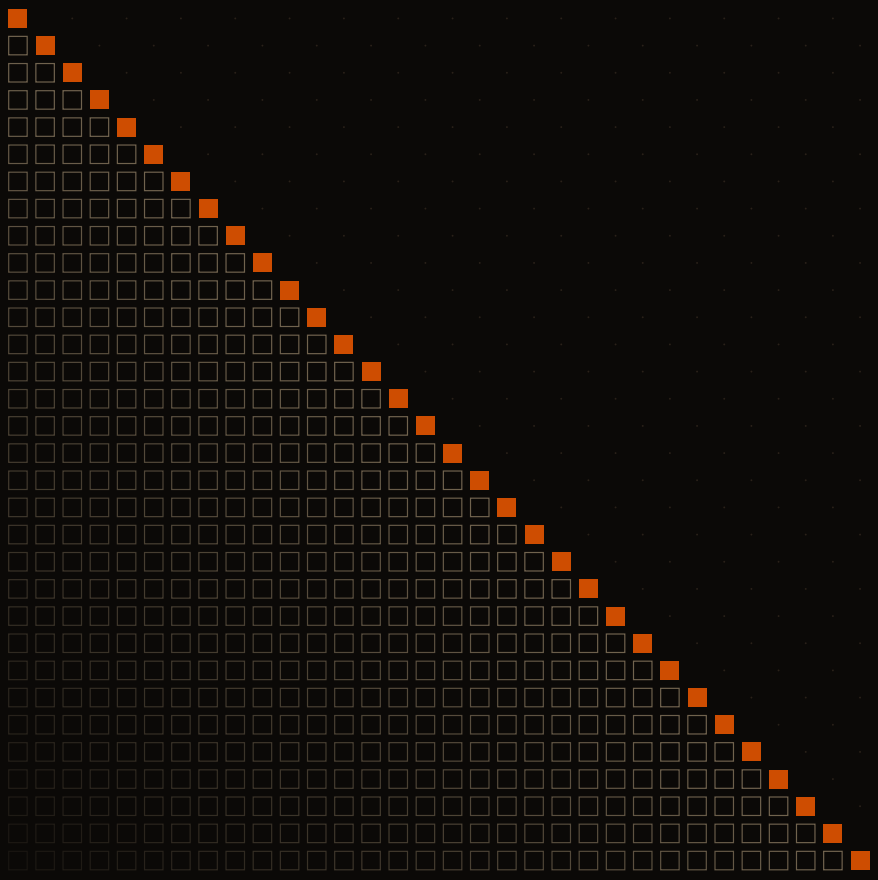


CAUSAL ATTENTION, N = 1024 · ONE SQUARE PER 32 × 32 TILE · NOTHING N × N WRITTEN DERIVED, NOT WRITTEN

Two Primitives Are Enough

Deriving flash attention, its backward pass, and whole-model kernels from affine views and monoid comprehensions

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Abstract

Every fast model today runs on kernels someone wrote by hand: one per operator, per chip, per fusion. Each one freezes choices about what to fuse, where data lives and how threads share work, and those choices are right for one shape on one device. We think kernels should be derived instead.

tang's tensor IR has two primitives. Views are affine maps from an iteration space to tensor coordinates. Comprehensions are reductions under a monoid whose body is a scalar expression graph. Everything else is either a rewrite or a schedule. Fusion is one substitution rule. Flash attention is not written: it falls out when the compiler recognizes online softmax as a monoid. Gradients of comprehensions are comprehensions, so the backward pass is derived by the same machinery. That includes a flash-style attention backward that writes no $n \times n$ intermediates on any backend (on tensor cores at full speed, and on WGSL and Metal for now at a cost we report); the concurrent superoptimizers we discuss derive only the forward. Precision and thread mapping are schedule fields that cannot change the answer, chosen per kernel by measurement on the device.

One graph lowers to WGSL, CUDA tensor cores, Metal simdgroup matrices, OpenCL, GLSL and C, from a desktop GPU and a laptop down to a humanoid robot's phone-class SoC. On an RTX 3090 a derived transformer block's forward pass at sequence length 512 runs 1.37× faster than PyTorch with TF32. Qwen3-8B decodes as a single IR graph at 55% of memory-bandwidth speed of light, measured on a shared GPU. There are no per-operator kernels: the one pattern-specific emitter, CUDA's flash kernel, matches the online-softmax monoid, not attention. We then hold the IR against a hand-tuned engine that serves an 80B mixture-of-experts model at 1,279 tok/s on one H200. The first layer we derived from it, the MoE expert GEMMs with int8 tensor cores expressed as schedule fields, runs 1.24× faster than the engine's hand kernels at batch 64 and trails by 8% at batch 1.

1 Introduction

A modern model has a few dozen distinct operators. A modern framework has thousands of kernels. Every (operator, backend) pair needs one. So does every fusion that pays: attention with its softmax, a matmul with its bias and activation, a norm with the projection after it. FlashAttention[1] is the canonical case. It is a large speedup that came from rewriting one operator by hand so its $n \times n$ score matrix never reaches memory. Its backward pass took a second hand-written kernel. Hopper took a third[2], and Blackwell's FP4 tensor cores are taking a fourth.

Hand kernels are forks. Each re-derives the same arithmetic and freezes a schedule that was right for one shape on one device, and we measured how often that schedule is wrong. On an Apple M4 Max, the fastest schedule for causal attention flipped between two runs of the same code: one run had $n = 512$ fastest as the naive graph with n^2 threads and $n = 1024$ fastest tiled, and a later run had the reverse (Table 2). The naive graph wins when its scores fit in cache and parallelism, not bandwidth, is the bottleneck, which depends on the size and even on the machine's power state. A 16×16 shared-memory matmul tile wins at $128 \times 576 \times 1536$ and loses at 1024^2. Rules of thumb tuned on the M4 picked the wrong matmul schedule on an RTX 3090. The winning schedule is a property of the graph, the chip and the size, not of the operator.

So we propose a different contract: write the model once, as math, and let the compiler derive every kernel. That works only if the IR is small enough to reason about and expressive enough that nothing important falls outside it. tang's has two primitives:

- **Views**: affine index maps that express broadcast, transpose, slicing, striding, padding and per-head column selection without touching memory.
- **Comprehensions**: $\text{out}[i] = \oplus_k \text{body}(x_0[v_0(i,k)], x_1[v_1(i,k)], \ldots)$, where body is a scalar expression graph and $\oplus$ is a monoid.

Contributions:

1. **An IR where kernels are theorems.** Fusion, flash attention, logsumexp and Welford variance are instances of two general rewrites (§3).
2. **Derived backward passes.** Reverse-mode differentiation maps comprehensions to comprehensions, including the online-softmax monoid. A proof procedure removes the stabilizing max's gradient, and with nested contractions this yields an attention backward with no $n \times n$ tensors (§4).
3. **Precision as a schedule.** An emulating interpreter predicts the error of a format choice before it runs (§5).
4. **One graph, six targets.** Lowerings from shared emitters, measured against hand kernels, PyTorch and the bandwidth limit on three chips (§6, §7).
5. **The bar, and a road to it.** A hand-tuned H200 engine for an 80B MoE model, and the specific IR extensions that would let tang derive every kernel in it (§8, §11).

2 Background: tang

tang is a `no_std` Rust library that started as a geometry and physics math crate. One `Scalar` trait covers `f32`, `f64` and dual numbers, so a simulation written once yields exact derivatives by changing a type parameter. `tang-expr` traces Rust closures into a scalar DAG of nine RISC operations (add, multiply, negate, reciprocal, square root, sine, atan2, 2^x, $\log_2$, plus a select), which it can differentiate symbolically, simplify and compile. The tensor IR uses that scalar DAG as its body language, so every comprehension body can be differentiated, simplified and compared structurally. Higher crates (`tang-gpu`, `tang-llm`, `tang-train`) run programs the IR produces.

3 The IR

3.1 Views

A view maps a point of an iteration space to a source tensor's coordinates with one affine form per source dimension, offset $+ \sum_j c_j \cdot \text{idx}_j$. Broadcast is a zero form, transpose a permutation, slice an offset, stride a coefficient. A form with two terms expresses "head h reads columns $64h + 2i$". Views compose by substitution. They are the only way a comprehension reads its inputs, so every layout decision is a view and every view can be analyzed.

A view may be guarded: an out-of-range read returns the monoid's identity (zero) rather than raising a shape error. Padded, dilated and transposed (polyphase) convolutions and overlap-add are guarded reads, with no padding operator. The gradient of a guarded read removes a coordinate rather than scattering. For $x[c, t + 3k - 9]$, $\partial x[c, \tau]$ reads $\partial y[o, \tau - 3k + 9]$ through a guard, so padding contributions never reach x.

Grouped-query attention's head / heads_per_group is not affine. The IR expresses heads as two dimensions, [groups, per_group], and the views do the rest. Dimensions can be fixed or symbolic (`Dim::Sym("seq")`). Construction, rewriting and differentiation work with symbols, and `bind` makes them concrete for scheduling and lowering.

A view may also gather: one source coordinate is data, read from an index tensor that is a sibling argument of the same comprehension, form $+ c \cdot \text{round}(\text{idx})$. MoE routing ($W[\text{sel}[t, k]]$), paged KV caches ($b_s \cdot \text{table}[\text{blk}] + \text{off}$) and embedding lookups become ordinary reads rather than special operators. The gradient of a gathered read is a scatter-add, written as a dense gated sum. Fusion survives: an affine map read through a gather fuses into it, and only nested gathers materialize.

3.2 Comprehensions and monoids

The only compute operation is

$$\text{out}[i_1, \ldots, i_p] = \bigoplus_{k_1, \ldots, k_r} \text{body}(x_0[v_0(i, k)], \ldots, x_m[v_m(i, k)])$$

A map is a comprehension with no reduce dimensions. Sum and max are monoids. Matmul is $\sum_k a[i, k] b[k, j]$. Attention scores, convolution and gather are the same operation with different views. The monoid's `combine` is itself a scalar graph, and

a comprehension may carry tuple-valued state. Online softmax is the monoid on (m, l, acc); arg-max is the monoid on $(\text{index}, \text{value})$.

3.3 Rewrites

Fusion is one rule. A map is a pure function of its inputs, so `substitute_maps` splices its body into every consumer and composes the views. Map→map, map→reduce and transpose→anything are the same case. The rule is cost-aware: a map is not recomputed inside a reduction unless it is cheap. Reductions are never substituted. They are inlined by a schedule decision, which separates algebra from scheduling. A row-major reshape that only merges dimensions is a view, not a copy.

Epilogues ride the reduction. A map that reads a reduction point for point (a bias, an activation, the acc $/l$ that finishes softmax) runs on the thread that holds the final accumulator.

Siblings share a pass. Two reductions over the same space and monoid with the same readers become one reduction under the product monoid (`merge_siblings`). The gate and up projections of a SwiGLU MLP become one tuple-valued matmul, the activation becomes its epilogue, and the normalized input is read once.

Flash attention is a rewrite. `stabilize_softmax` finds $m = \max_j s_j$ and every $\sum_j g \cdot \exp(s_j - m)$ over the same canonical s, with masks and scales allowed, and replaces them with one reduction under the online-softmax monoid. The value column of $\sum_j p_j v_j$ becomes a point dimension of the fused comprehension. Nothing $n \times n$ is materialized. Logsumexp, Welford variance and top-k are the same rewrite with different monoids. Figure 1 shows the attention case.

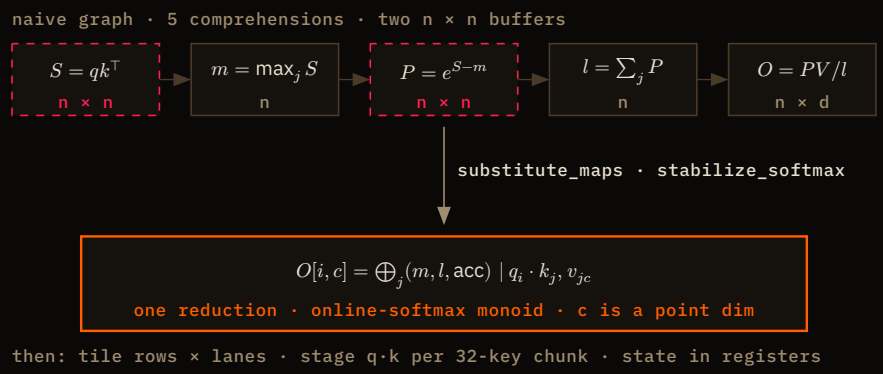


FIG 1 The rewrite that derives flash attention. The naive graph writes the scores and probabilities as $n \times n$ buffers (dashed). Fusion splices the maps into their readers; `stabilize_softmax` merges the max and both exponential sums into one reduction whose state is (m, l, acc). No attention-specific code is involved; scheduling then finds the tiled kernel.

The rewrite is not specific to plain attention. Anything that changes the score or the mask leaves the monoid structure alone, so it derives the same way. Table 1 derives six variants.[1]

[1] The web edition derives these in the reader's browser and runs them through WebGPU: muni.works/research/tang.

TABLE 1 Six attention variants as typed into the lab, $n = 1024$, $d = 64$. Each cell is kernels · $n \times n$ buffers, as written → derived (WGSL). The forward always becomes one kernel with nothing $n \times n$; with nested contractions (§4) so does the backward, in 7 kernels. Generated by `tang-derive-wasm`'s `gallery` example.

VARIANT	SCORE	MASK	FWD KERNELS · N×N	FWD+BWD KERNELS · N×N
Causal	`s`	`j <= i`	11 · 5 → **1 · 0**	30 · 11 → **7 · 0**
Sliding window	`s`	`j <= i && i - j < 128`	11 · 5 → **1 · 0**	30 · 11 → **7 · 0**
ALiBi	`s - 0.0625*(i - j)`	`j <= i`	11 · 5 → **1 · 0**	30 · 11 → **7 · 0**
Soft cap	`30*tanh(s/30)`	`j <= i`	11 · 5 → **1 · 0**	30 · 11 → **7 · 0**
Prefix LM	`s`	`j <= i || j < 64`	11 · 5 → **1 · 0**	30 · 11 → **7 · 0**
Bidirectional	`s`	—	10 · 4 → **1 · 0**	28 · 9 → **7 · 0**

3.4 Schedules

A comprehension says what to compute. Schedule fields say how, and they cannot change the answer:

- `inline`: recompute inside the consumer instead of writing a buffer.
- `split`: one workgroup per point, with threads striding the reduce space and tree-combining. This is valid for any monoid because combine is associative, including online softmax.
- `tile`, `tile_2d`: blocks of threads, each owning a register tile of points. The emitter tracks every address as an affine form over loop variables. Anything that does not depend on the lane axis is staged in workgroup memory once per reduce chunk. That covers buffer reads (K and V tiles, matmul operand tiles) and inlined sub-computations (the $q \cdot k$ scores).
- `precision`: storage and accumulation formats (§5).

`schedule_auto` encodes three rules distilled from measurement. `autotune_kernels` times each candidate mapping per materialized node on the device, using the graph's real inputs, and caches the winner by device and kernel source. With substitution and the online-softmax rewrite, the staging rule produces flash attention's real structure: rows × lanes per workgroup, cooperative score tiles, and online softmax in registers.

4 Gradients are comprehensions

For $y = \oplus_k \text{body}(x[v(i,k)])$, the gradient with respect to x through a pick view is found by renaming coordinates. The coordinates that v picks become x's output dimensions, the rest become reduce dimensions, and every other view is pushed through the same renaming. The body is differentiated by `tang-expr`'s symbolic `diff`. A dimension read through any other form (an offset, stride, diagonal or broadcast) stays an output dimension, and the body is gated by $[c = \text{form}]$ through `iota`, a node that turns coordinates into values. Max gates its term to the arg-max. The backward graph is an ordinary graph, so the same passes optimize it. Rotary embedding and GQA gradients come from the gated scatter with no special handling.

The online-softmax monoid has its own rule. With the forward's m and accumulators read back, $\partial \mathrm{acc}_i/\partial s_j = g_i \exp(s_j - m) - \mathrm{acc}_i\,[s_j = m]$, so the flash-fused graph differentiates and its gradients equal the naive graph's.

A stabilizing max has no gradient. Softmax is invariant to shifting its scores, so the $[s = m]$ terms sum to zero, but only in exact arithmetic. While those terms exist, the max must be materialized and masked tiles cannot be skipped. `backward` drops them when it can prove the loss does not depend on m. The proof tracks how each node downstream of m scales with $t = \exp(-m)$. In softmax, $p_j = \exp(s_j - m) = t \cdot \exp(s_j)$ has degree 1 in t, so does $l = \sum_j p_j$, and so does $\mathrm{acc} = \sum_j p_j v_j$; their ratio $o = \mathrm{acc}\ /l$ has degree 0, a function of the scores alone. In general a node is either $t^k h$ or $a + \kappa m$ with h, a free of m, combined by exact identities.[2] If every path from m ends at a degree-0 node, $\partial L/\partial m \equiv 0$ and the gradient is not built. The f64 evaluation of a training step is unchanged (largest absolute difference 2.7×10^{-19}).

Flash backward is nesting. `hoist_invariant_factors` distributes sums ($\sum_c \tilde{P} \cdot \mathrm{dO} \cdot V \to \tilde{P} \cdot \sum_c \mathrm{dO} \cdot V$). `schedule_nested_contractions` then inlines a short contraction (one reduce dimension of at most 128) into every tile that reads it. The attention backward reads $S = QK^\top$ and $\mathrm{dP} = \mathrm{dO}\ V^\top$ only through $\tilde{P}$ and dS, which the dQ, dK and dV tiles recompute per staged block on tensor cores. On CUDA this removed S, $\tilde{P}$, dP, the max over S and three correction kernels from a transformer training step, reducing 63 kernels to 49 and the summed best-per-kernel time from 1.80 to 1.22 ms at sequence length 512 (RTX 3090). WGSL and Metal now nest the same contractions with scalar dot products in workgroup memory, so no backend writes anything $n \times n$. That costs speed for now: forward plus backward runs at 0.64–0.78× the materialized schedule on Metal and 0.22–0.27× under Vulkan on a 3090, until the nested dots are register-blocked the way the `mma` path blocks them.

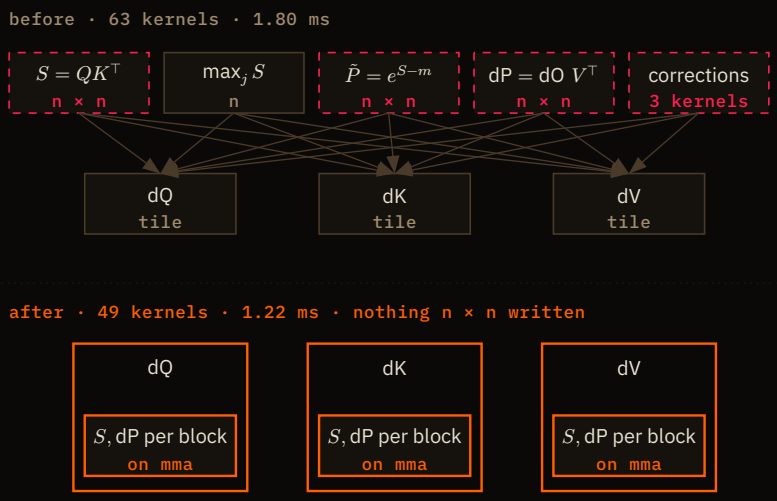


FIG 2 The attention backward of a training step, before and after. Before, the scores, probabilities and dP are written as $n \times n$ buffers (dashed) and read by three gradient tiles. After the stabilizing max loses its gradient and sums distribute, the short contractions S and dP are recomputed per staged block inside each tile, on tensor cores. CUDA, RTX 3090, seq 512; summed best-per-kernel times.

[2] $t^a t^b = t^{a+b}$; $\sum_j t h_j = t \sum_j h_j$ when t's row does not depend on j; $2^{a+\kappa m} = 2^a t^{-\kappa \ln 2}$.

5 Precision is a schedule

f16 is a different way to run the same program, not a different program. Each node has a storage format, applied where a value is written to memory, and an accumulation format, applied to a reduction's running state after each combine. The graph stays exact math.

The interpreter predicts the error. `eval` is pure f64. `eval_emulated` rounds every stored value and every accumulator step to its scheduled format (round to nearest even, subnormals, overflow), with body arithmetic in f64. An `ErrorBudget` accepts or rejects a schedule before it runs. Emulation matches the GPU's f16 kernels to 1 ulp when the reduction order is the same. Reduction order matters: AdaIN's variance over $T = 2000$ with an f16 accumulator has error 2.8×10^{-4} under a 256-thread split and 1.3×10^{-2} with one thread per row, while the sequential emulation predicts 7.9×10^{-3}. The reduction order is part of the error: emulation models a sequential fold, which a tree-combining split beats and one thread per row does slightly worse than. `tune_precision` lowers buffers to half precision one move at a time and keeps a move only if the program stays within budget and gets faster.

6 Lowering

Shared emitters produce the point, split and tiled schedules, register tiles, guarded views, epilogue fusion and tuple monoids once. A dialect switch then handles buffer access and surface syntax.

- **WGSL** (portable default, through wgpu on Metal, Vulkan and DX12). f16 storage is packed two per `u32`, because the toolchain rejects `enable f16`.
- **CUDA.** Matmul-shaped tiles run on `mma.sync` (TF32 m16n8k8, or f16/bf16 m16n8k16 when the operands are half precision) with `cp.async` staging rings. Block shapes are tuning candidates. `cuda_flash_kernel` recognizes the online-softmax comprehension and keeps O, the row max and the partial sums in `mma` registers. The score accumulator is reused as the A operand of $O += PV$ after permuting keys within each 8-wide step, which is legal because the reduction is associative.
- **Metal (MSL).** Native `half` and `bfloat`. Contractions with f16 operands run on `simdgroup_half8x8` fragments. Reduce dimensions are flattened, so convolution becomes implicit GEMM, with dilated inputs staged once per chunk as a zero-padded, phase-de-interleaved window.
- **OpenCL and GLSL** for Adreno GPUs that have neither CUDA nor Metal. On one Adreno driver, WGSL is routed through GLSL because the direct SPIR-V path was miscompiled.
- **C**, for batch-1 control loops where a GPU round trip costs more than the arithmetic. A comprehension already is a loop nest, so `to_c` emits it directly with a vectorizable innermost point dimension, and `tang_run()` performs no allocation.

`Program::tuned` times each kernel's native lowering against the portable one and keeps the faster. Kernel sources are canonicalized, so the same kernel at another layer is the same string and compiles once.

7 Evaluation

Setup. Apple M4 Max (Metal), NVIDIA RTX 3090 (Vulkan and CUDA 12), and a Qualcomm QCS8550 in a Booster K1 humanoid robot (Adreno 740, Cortex-X3). PyTorch 2.10 + cu128 with CUDA events. Unless stated otherwise, tang times are kernel-only. Several runs shared the GPU with other tenants. We mark those, and report minima rather than means, because contention only adds time. Every derived result is checked against an f64 interpreter; agreement is 10^{-6} to 10^{-4} relative, depending on format.

7.1 Derived vs hand-written kernels

tang shipped hand-written WGSL kernels before the IR existed, which makes a controlled comparison possible: same runtime, same chip, same inputs. Table 2 is one consistent run on the M4 Max, including about 1.8 ms of submit and readback per measurement.

TABLE 2 Hand-written vs derived on M4 Max (Metal via WGSL). Every time includes about 1.8 ms of submit and readback, which dominates the small ops. The hand training step ran its attention backward on the CPU, since no GPU version existed, so that row shows coverage, not kernel quality.

OP	HAND	DERIVED	SCHEDULE
causal attention $n = 1024$	10.9 ms	**3.31**	naive (n^2 threads)
causal attention $n = 512$	4.91	**2.02**	auto (tiled)
matmul 1024^2	6.68	5.49	auto
rmsnorm 2048^2	2.52	2.70	split
transformer block, seq 512	23.9	**10.1**	per-kernel autotune
decode block, q 1 / kv 1024	8.80	**3.28**	per-kernel autotune
training step, seq 128	179	**7.96**	auto

Every hand kernel is beaten or matched to within 7% (rmsnorm), leaving aside the training step. The schedules that win differ across operators and sizes. On the 3090 under Vulkan, the derived transformer block first lost to the hand path. Three chip-agnostic fixes followed: cost-aware substitution, keeping reductions schedulable under their epilogues, and a bank-conflict-free tile layout. After them, the autotuned block won there too (2.5 vs 3.1 ms at seq 128, 6.2 vs 10.4 ms at seq 512).

Figure 3 shows the same thing through WebGPU in a browser on the M4 Max. At $n = 1024$ the graph as written beats every derived schedule, and the derived schedules alone span 29× by thread mapping. A compiler that fixed one schedule would be wrong somewhere, so tang times the candidates and keeps the winner.

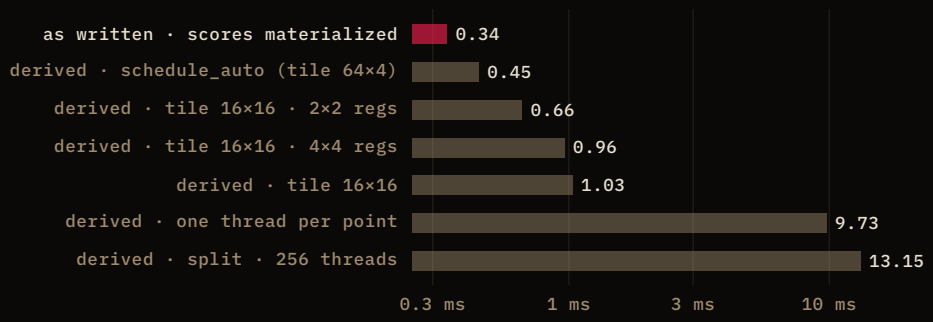


FIG 3 Causal attention, $n = 1024$, $d = 64$: the schedules the post's lab timed through WebGPU in Chrome on an M4 Max (2026-11-06, log scale). Red: the graph as written. All outputs match the f64 interpreter to 1.3×10^{-7} (checked at $n = 128$).

7.2 Against PyTorch

Table 3 tracks a 9-head GQA transformer block ($d = 576$, ff $= 1536$, rotary embeddings, SwiGLU) on the RTX 3090 as the backends matured. Through WGSL, which has no access to tensor cores, derived kernels stayed 2.5–4× behind PyTorch. With the CUDA lowering, flash attention on `mma.sync` and the backward passes of §4, the forward block passes PyTorch (Figure 4).

TABLE 3 Transformer block on the RTX 3090. The CUDA column and PyTorch were measured in interleaved rounds under `flock`, best of six; times in ms; PyTorch's 4-layer step uses fused AdamW. The WGSL columns come from earlier runs and are shown to make the trajectory visible.

BLOCK, SEQ	WGSL FIRST	WGSL FUSED	CUDA NOW	TORCH FP32/TF32
fwd 128	1.97	1.06	**0.178**	0.306 / 0.286
fwd 512	4.41	3.34	**0.328**	0.602 / 0.449
fwd+bwd 128	5.0	3.15	**0.552**	1.196 / —
fwd+bwd 512	10.1	8.19	**1.467**	1.719 / —
4 layers, step 128	—	—	**3.92**	4.40

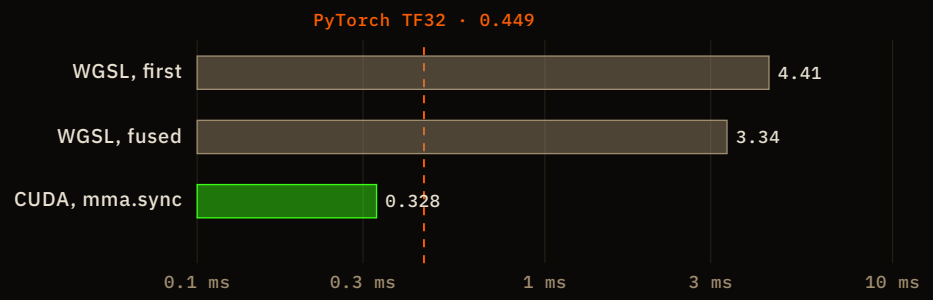


FIG 4 Transformer block forward, seq 512, RTX 3090, log scale. The same derived graph under three generations of lowering; only the last reaches tensor cores. Dashed: PyTorch with TF32.

On the CUDA path, the derived inference attention kernel is 0.032 ms at seq 512, down from 0.178 ms before the `mma.sync` rewrite. The seq-512 forward block is 1.37× faster than PyTorch with TF32. Forward plus backward runs 2.2× faster than PyTorch fp32 at seq 128 and 1.17× at seq 512, and a four-layer training step 1.12× faster than PyTorch with fused AdamW. We do not claim a general win over cuBLAS: a lone fp32 matmul at 1024^2 through WGSL reaches 7.6 TFLOPS against cuBLAS's roughly 19.

The gain comes from fusion across operator boundaries that a library of separately tuned operators cannot do.

7.3 Attention variants against FlexAttention and FA2

We ran the six attention variants from §3.3 on the RTX 3090 against FlexAttention[3] (`torch.compile`, fp32 with TF32 and bf16) and FlashAttention-2 in bf16 where it supports the mask. Flashlight and EqiForge publish no code. Every row was checked against an f64 reference; tang is pure fp32 or TF32.

TABLE 4 Attention variants on an RTX 3090 (shared; small rows carry up to ±2× noise). FA2 runs only causal and bidirectional masks. The full sweep, d = 64 and 128 with n from 1k to 8k, is in the repository.

D = 64 UNLESS NOTED, MS	TANG	FLEX FP32	FLEX BF16	FA2 BF16
soft cap, fwd, n = 2k	**0.075**	0.254	0.172	—
ALiBi, fwd, n = 2k	**0.060**	0.171	0.094	—
causal, fwd, n = 2k	0.060	0.171	0.090	**0.023**
causal, fwd, n = 8k	0.888	0.635	0.326	**0.177**
sliding window 128, fwd, n = 8k	0.911	0.037	**0.034**	—
causal, fwd + bwd, n = 4k	1.81	0.968	**0.420**	0.469
causal, fwd, n = 4k, d = 128	17.1	fails	0.318	**0.112**

The results split cleanly. Where a variant changes the score (soft cap, ALiBi, prefix LM), the derived forward is 1.4–3.4× faster than FlexAttention in fp32 up to $n = 4$k and beats it in bf16 up to $n = 2$k: the score modifier simply becomes part of the fused body. tang loses in three places, each a missing schedule rather than missing algebra. FA2 runs bf16 tensor cores where tang's tile is TF32. A sliding window is 25× slower because tang computes masked tiles that FlexAttention's block mask skips. And at $d = 128$ no tensor-core tile is chosen for the forward. From $n = 2$k the backward is 2.5–7× behind FlexAttention bf16, and 50× with a sliding window.

7.4 A whole model as one graph

Qwen3-8B with MLX 4-bit weights runs as one IR graph per decode step, lowered by `to_cuda_decode` into 506 kernels and captured as a CUDA graph. Nibble unpacking and the $q \cdot s + b$ dequantization are a map the GEMV reads. That map is inlined and evaluated in registers as weights stream in. Figure 5 shows where the remaining time goes. The KV-cache update is a `select` over a runtime position, which the lowering turns into loop bounds, so a step touches only live cache rows.

TABLE 5 Measured while another job held about 98% of SM time, so the true numbers are likely higher. Teacher-forced on 64 greedy tokens against an exactly dequantized f32 CPU reference: max logit error 1.7×10^{-4}, with 64/64 arg-max agreement.

QWEN3-8B, 3090	DECODE, SHORT CTX	DECODE, 4K CTX	PREFILL 512
speed of light (936 GB/s)	220 tok/s	192 tok/s	—
tang IR, device step	**122 tok/s** (55%)	**104 tok/s** (54%)	**774 tok/s**

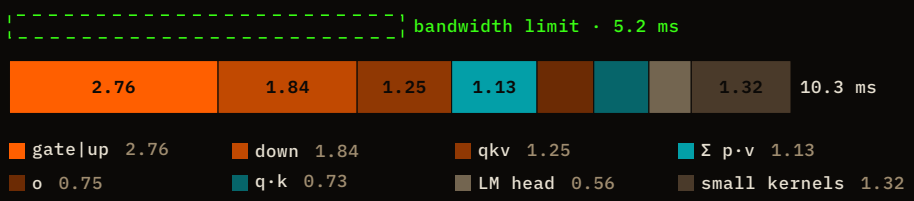


FIG 5 Where a Qwen3-8B decode step goes at 4k context: the sum of isolated kernel launches (10.3 ms; the captured graph runs in 9.6). The large GEMVs dominate; about 400 small reductions and maps add 1.3 ms of mostly fixed launch cost. Dashed: the time to stream the weights and KV cache once at 936 GB/s.

The same graph builder lowers to Metal. There, prefill and decode programs share one copy of the weights through buffer binding, prefill can start from a nonzero offset (300 + 200 tokens is bit-identical to 500), and the next token is chosen on the device by an arg-max monoid, so decoding reads back 8 bytes per token.

7.5 Data-dependent addressing

A top-k MoE FFN builds as one graph: router GEMM, top-k as k arg-max rounds, softmax weights, SwiGLU experts reading $W[\mathrm{sel}[t,k]]$ through gather views, then the weighted combine. Top-k is exact, and outputs match the f64 interpreter to 4×10^{-6} on Metal and 4×10^{-5} on CUDA (the router runs as a TF32 tile).

TABLE 6 Top-k MoE at T = 8 tokens, D = 2048, E = 128 experts, k = 10, F = 512 (1 GB of weight reads). The data-dependent address costs nothing measurable against the same expert math over weights gathered into contiguous buffers beforehand. 378 GB/s is 40% of the 3090's peak: per-point GEMV reads are uncoalesced, and a warp-per-row schedule for gathered contractions is next.

EXPERT GEMMS + COMBINE	M4 MAX	RTX 3090 (CUDA)
gathered (one graph)	**2.17 ms** (464 GB/s)	**2.66 ms** (378 GB/s)
weights pre-gathered on the host, tiled	2.89	3.55
weights pre-gathered, one thread per point	2.91	3.77

7.6 Edge hardware

On the K1's Adreno 740, a derived matmul reaches 322 GFLOP/s at 1024^3, which is 21% of the measured fp32 peak, and a full transformer forward pass plus all seven weight gradients runs in 19.1 ms. Rules tuned for desktop GPUs under-fill the Adreno's six compute units. The best schedule there (an 8×8 register tile with 8-deep chunks) is 1.45× better than `schedule_auto`'s choice, which again argues for search over remembered rules. Through `to_c` on one Cortex-X3 core, a batch-1 GRU policy step (6 contractions plus gate math, fused into one kernel) takes 22.5 µs median and a locomotion MLP takes 6.7 µs, both within 6×10^{-7} of f64.

8 The bar, and the first layer over it

Everything above is derived. To show where derivation has to go next, we report the other half of tang: `tang-llm`'s Flash engine, which serves Qwen3-Coder-Next (80B parameters, 3B active, 4-bit GGUF) on one NVIDIA H200 with kernels written by hand. Table 7's numbers are not results of the IR; they are the target it has to reach.

TABLE 7 The bar. tang-llm on a dedicated H200 (tang-llm commit `fcfbb9d7`). Its MoE layers run a hand-written int8 tensor-core kernel that reads activations in B-fragment order, which cut the B = 64 step from 38.9 to 35.2 ms. Lossless speculation uses a trained MTP head (3.15 tokens per window).

H200, QWEN3-CODER-NEXT 80B-A3B	MEASURED
single-stream decode, no drafting	234 tok/s
with speculative drafting (MTP / suffix)	319–439 tok/s
prefill, 1k → 8k tokens	6,574 → 7,548 tok/s
batched decode step, B = 64	1,814 tok/s
serving, 64 concurrent decode-heavy streams	1,279 tok/s
serving at 64k context, batch 48 (production shape)	1,155 tok/s

The hand engine also shows why the contract matters. Ported to GLM-5.3, the same MoE kernel is correct but 1–2.5× slower below 64 tokens per expert and only 8–20% faster at 64. A kernel tuned for one model's expert shapes is a point in the schedule space, and on the next model it is the wrong point. It is one data point, but from a production engine, and it is §1′s claim.

Deriving this engine needed three things the IR lacked: a gather view for routing tokens to experts, integer tensor-core fragments with fused dequantization, and Hopper's asynchronous copy and warpgroup MMA as schedule fields. The first two now exist, and with them we derived one full MoE layer of the engine as a single graph: router, top-k, expert-sorted gather, the gate|up and down expert GEMMs on GGUF Q4_K/Q5_K weights, SwiGLU and the weighted combine. The int8 path is schedule fields only: GGUF blocks as an input storage format, int8 activations quantized by their writer, and an `IntMma` field that stages activations in fragment order for `m16n8k32` tensor cores. There is no new operator.

TABLE 8 One Qwen3-Coder-Next MoE layer (hidden 2048, 512 experts, top-10, FFN 512), derived vs the engine's own kernels on the same bytes, on an idle H200. Medians of 100 launches with L2 flushed. Outputs agree to 3×10^{-7}. Weights are random valid k-quant blocks.

EXPERT HALF OF ONE LAYER, H200	DERIVED FROM THE IR	ENGINE, HAND-WRITTEN
batch 1	0.048 ms	**0.045 ms**
batch 8	**0.098**	0.126
batch 64	**0.270**	0.336
batch 64, gate|up / down	**0.158** / **0.089**	0.180 / 0.124

At batch 8 and 64 the derived kernels beat the hand ones, which took a dedicated tuning effort to push past a 2 TB/s floor. Fragment-order staging alone is worth about 20% on gate|up. At batch 1 the hand per-pair GEMV still wins, because a narrow one-pair-over-many-warps schedule is not yet expressible. The derived routing is the other gap: top-k as k arg-max rounds costs 27 small kernels (0.35–0.6 ms) where the engine uses one.

9 Verification

Speed claims depend on every kernel being correct. `tang-verify` runs each GPU kernel against its CPU oracle on the same input, starting from the device's own state. Before each kernel runs, it fills the kernel's outputs with NaN, so unwritten memory cannot pass as a plausible 0. When a check fails, it identifies the responsible kernel by provenance: whose correct values ended up in this buffer. It was built after four structural bugs in a hand-written engine slipped past every unit test. One was a byte offset added where a float offset was meant, which wrote k's values over q. Derived kernels are less exposed to that class of bug, because addresses come from affine forms rather than pointer arithmetic. The IR path also has a structural oracle: the f64 interpreter runs the identical graph.

10 Related work

Schedules. Halide[4] separated algorithm from schedule. TVM[5] and Ansor[6] searched the schedule space with learned cost models, and Tensor Comprehensions[7] used polyhedral comprehensions. tang inherits the separation and the search.

Deriving fused kernels. The closest work is concurrent. Mirage[8] superoptimizes multi-level μGraphs with probabilistic equivalence checks. EqiForge[9] uses equality saturation to derive FlashAttention-style forward kernels. Flashlight[10] generates flash-style kernels for arbitrary attention variants inside TorchInductor. tang differs on three axes. Its rewrite is deterministic, keyed on the monoid structure rather than found by search. It derives the backward pass, including the gradient of the online-softmax monoid and the elimination of the stabilizing max. And it lowers to six targets, not one. Neither Flashlight nor EqiForge publishes code, so we measure against FlexAttention and FA2 instead (§7.3).

Tile languages and hand kernels. Triton[11], ThunderKittens[12] and TileLang[13] make kernels easier to write. FlashAttention[1,2,14] is the hand-derived frontier. Online softmax is due to Milakov and Gimelshein[15]. tang aims to make writing kernels unnecessary, and treats these kernels as the targets derived code must reach.

Generated kernels. LLM agents now write CUDA, but KernelBench-Verified[16] shows frontier models reward-hacking benchmarks by hard-coding test values. A derived kernel is checked by construction against an f64 interpreter of the same graph.

11 The road to every kernel

Each gap below is a missing primitive or schedule field, not a missing kernel. We list what each one unlocks.

- **Faster gathers.** Gather views exist (§3, Table 6). Gathered contractions still run one thread per point; a warp-per-row schedule, a segmented-sum backward and expert-major dispatch (which needs a scan) are next.
- **Hopper and Blackwell as schedule fields.** TMA bulk copies, warpgroup MMA, warp specialization and FP8/FP4 fragments are the asynchrony FlashAttention-3 hand-codes[2]. In tang they belong next to `tile` and `precision`, so the derived flash

forward and backward meet FA3 on Hopper rather than describing it. bf16 tensor-core tiles everywhere, including $d = 128$, come with them.

- **Mask-aware tile skipping everywhere.** The CUDA flash kernel already skips key tiles an affine mask covers. The WGSL and generic CUDA tiles do not, which is why a sliding window loses 25× (Table 4).
- **Register-blocked nesting.** The scalar nested dots that make the WGSL and Metal backward $n \times n$-free (§4) need blocking into small GEMMs, on simdgroup matrices on Metal, to be free.
- **The rest of the MoE layer.** A warp-level top-k in place of k arg-max rounds, the engine's norm, SiLU and combine fusions, and a narrow batch-1 schedule close the remaining gaps in Table 8.
- **Scan monoids.** State-space and gated-delta recurrences are associative scans over a matrix monoid. ScanWeaver[17] makes scan a first-class compiler abstraction; tang's monoids are already the right shape for it, and its backward rule would come with it.
- **Launch fusion.** About 400 of a Qwen decode step's 506 kernels are small reductions and maps that cost 1.3 ms in launches. Persistent megakernels[18] are a schedule over the whole program, not a new operator.
- **A learned cost model.** `schedule_auto` is three measured rules. Every autotuning run already writes (graph, chip, schedule, time) to a cache, and that cache is training data.
- **Device-to-device placement.** Sharded reductions joined by an all-reduce are already a rewrite (the all-reduce is a comprehension over the monoid's combine). Only the transport is missing.
- **Quiet measurements.** Several runs here were taken on shared GPUs and are reported as minima. Every table should be re-run on idle hardware.

12 Conclusion

Two primitives, affine views and monoid comprehensions, are enough to express a transformer, its gradient and its quantized inference. Fusion, flash attention and the flash backward become rewrites and schedule choices, not kernels. Because the schedule is searched per chip, the same source matches or beats hand kernels on Metal and Vulkan, runs on an embedded Adreno, and beats PyTorch on a transformer block's forward pass on CUDA.

The hand-written kernel does not disappear in this design. It becomes an intrinsic: a known-fast implementation of one comprehension shape on one chip, which the compiler uses when it wins and replaces when it doesn't. The last hand-tuned engine we run serves an 80B model on an H200, and the first of its layers we derived already beats the kernels written for it. The goal is a compiler that derives every kernel in it, for every chip we will ever run on.

> **ASIDE · REPRODUCIBILITY**
> Every number in this paper comes from `BENCHMARKS.md` in the tang repository (snapshot in `data/`), which records the command, date and machine state for each measurement. The IR is in `crates/tang-compile/src/tensor*.rs`; its design notes are in `docs/tensor-ir.md`.

References

1. Dao, T., Fu, D. Y., Ermon, S., Rudra, A. & Ré, C. FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. in Advances in Neural Information Processing Systems (2022).
2. Shah, J. et al. FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision. arXiv preprint arXiv:2407.08608 (2024).
3. Dong, J., Feng, B., Guessous, D., Liang, Y. & He, H. Flex Attention: A Programming Model for Generating Optimized Attention Kernels. arXiv preprint arXiv:2412.05496 (2024).
4. Ragan-Kelley, J. et al. Halide: A Language and Compiler for Optimizing Parallelism, Locality, and Recomputation in Image Processing Pipelines. in PLDI (2013).
5. Chen, T., Moreau, T., Jiang, Z. & others. TVM: An Automated End-to-End Optimizing Compiler for Deep Learning. in OSDI (2018).
6. Zheng, L., Jia, C., Sun, M. & others. Ansor: Generating High-Performance Tensor Programs for Deep Learning. in OSDI (2020).
7. Vasilache, N., Zinenko, O., Theodoridis, T. & others. Tensor Comprehensions: Framework-Agnostic High-Performance Machine Learning Abstractions. arXiv preprint arXiv:1802.04730 (2018).
8. Wu, M., Cheng, X., Liu, S. & others. Mirage: A Multi-Level Superoptimizer for Tensor Programs. arXiv preprint arXiv:2405.05751 (2024).
9. Zhan, Q., Hu, X., Xia, X. & Li, S. Unleashing the Power of Equality Saturation for Tensor Program Superoptimization. arXiv preprint arXiv:2609.12330 (2026).
10. You, B., Wang, I., Mustafaoglu, Z. S., Jangda, A. & others. Flashlight: PyTorch Compiler Extensions to Accelerate Attention Variants. arXiv preprint arXiv:2511.02043 (2025).
11. Tillet, P., Kung, H. T. & Cox, D. Triton: An Intermediate Language and Compiler for Tiled Neural Network Computations. in MAPL (2019).
12. Spector, B. F., Arora, S., Singhal, A., Fu, D. Y. & Ré, C. ThunderKittens: Simple, Fast, and Adorable AI Kernels. arXiv preprint arXiv:2410.20399 (2024).
13. Wang, L. & others. TileLang: A Composable Tiled Programming Model for AI Systems. arXiv preprint arXiv:2504.17577 (2025).
14. Dao, T. FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning. arXiv preprint arXiv:2307.08691 (2023).

15. Milakov, M. & Gimelshein, N. Online normalizer calculation for softmax. arXiv preprint arXiv:1805.02867 (2018).

16. Zhang, Y., Yu, P., Wang, J. & others. KernelBench-Verified: Do LLM-Generated Kernels Actually Beat PyTorch? arXiv preprint arXiv:2607.16241 (2026).

17. Wu, Q. & Zolnikov, P. ScanWeaver: Compiler-Driven Parallelization of Affine Recurrences via Associative Scan Lowering. arXiv preprint arXiv:2606.00601 (2026).

18. Jin, H., Hou, B., Wang, G., Lai, R. & others. Event Tensor: A Unified Abstraction for Compiling Dynamic Megakernel. arXiv preprint arXiv:2604.13327 (2026).